

1.4c

The Brain

Damage Response & Brain Hemispheres

How can our brain's heal

Most brain-damage effects are due to 2 facts:

Severed brain & spinal cord neurons don't regenerate
Some brain functions seem preassigned to specific areas
ex. damage $\rightarrow$ temp. lobe $\Rightarrow$ can't recognize faces.

Some neural tissue can reorganize in response to damage

Blind person uses finger to read
 $\Rightarrow$ brain area for that finger $\uparrow$, touch sense invades visual cortex

Deaf ppl. who learned sign lang before others may have better peripheral & motion detection vision
Their temp lobe never received stimulation, so it looks for other signals, like visual ones

If a tumor growing in the left hemisphere disrupts language, (which is concentrated in the left), the right may compensate.

Finger amputated $\Rightarrow$  finger sensitivity $\uparrow$

The Brain often repairs thru reorganization
Researchers are debating whether it can be helped by producing new neurons: neurogenesis

Researchers have found baby neurons deep in the brains of adult mice, birds, monkeys & humans

These baby neurons may then form connections with neighbouring neurons.

Stem cells have been discovered in the human embryo
Maybe, injecting new stem cells into a damaged brain will cause the new stem cells to become replacement neurons.

Natural neurogenesis promoters: Exercise, Sleep, nonstressful but stimulating environments.

Whats the difference between our hemispheres?

left & right hemispheres have different functions

Lhem accidents, strokes & tumors can impair reading, writing, speaking, arithmetic reasoning, & understanding.

Similar Rhem damage has less intense effects

Corpus Callosum: wide band of axon fibers connecting hemispheres & carrying messages between them

In 1960s, researchers severed the CC in epileptic patients.

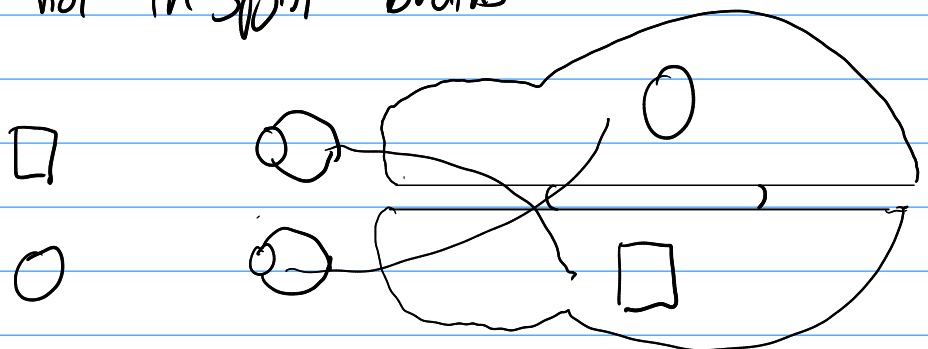
The patients with these split brains didn't have seizures any more. Personality, intellect ~ same.

Each eye receives sensory info from entire visual field.

Info from left $\frac{1}{2}$ (FOV) $\rightarrow$ Rhem

Info from right $\frac{1}{2}$ (POV) $\rightarrow$ Lhem (which controls speech)

Info received by either hem quickly sent to other through CC, but not in split brains



In split brains researchers could quiz each hemisphere separately by getting a participant to stare at a dot while flashing stimulus to right or left.

Participants were shown

HE • ART

left half right half

When asked to SAY what they'd seen, they said art. (Speaking controlled by Lhem)

But when asked to point on a paper with their left hand (controlled by Rhem) what they saw, they pointed at He.

Given an opportunity to express itself, each hemisphere indicated what it had seen.

Rhem intuitively knew what it couldn't verbally report

When a picture of a spoon was flashed to Rhem, participants couldn't say what they saw, but could identify what they had seen by feeling an assortment of hidden objects with their left hand.

Some with split brains have had to deal with an independent left hand, which would unbutton a shirt the right was buttoning.

With a split brain, participants can simultaneously draw, copying different figures with each hand.

Lhem, does mental gymnastics to rationalize actions it does not understand.

Patient follows order to walk sent to Rhem, Lhem does not know why. When asked, the patient doesn't say 'I don't know', Lhem improvises 'I'm going into the house to get a coke'.

Lhem: an 'interpreter' that instantly constructs explanations.

Brain runs on autopilot: act first, then explains itself.

What about intact brains?

When one performs a perceptual task, brain activity is increased in Rhem.

When one speaks or does a calculation, brain activity ↑ in Lhem.

For the brain, language is language whether spoken or signed.

just as hearing ppl use Lhem, to process spoken language
deaf ppl use Lhem to process sign language

Although Lhem is good with quick, literal, interpretations of language,

Rhem:

- good w/ inferences; eg. "what word goes wth boat, summer, grand"
- helps modulate speech for clarity:
"Lets eat, Grandpa" vs "Lets eat Grandpa"
- helps w/ self-awareness

Personal Response:

I generally find that qualities I excel in are associated with Lhem. This could indicate that my Lhem may have significantly more neural connections than Rhem

MCQ:

1. c 2. b 3. a

4. b 5. a 6. b.